

The Modular Hop Is CHM, Not Bloch: a Complete Covering of the Ball

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Abstract

A guess: virtual modes live on a Bloch 3-sphere, so the modular hop should track a Bloch vector, not the CHM weight. The testable piece is scored on *every* nearest-neighbour dimer in the ball.

For the 2×2 correlator of a dimer, $r_x = 2C_{ij}$, $r_z = C_{ii} - C_{jj}$. $N = 16$, $R = 5$: 1302 dimers. $N = 12$, $R = 4$: 624 dimers. Auditor $N = 14$, $R = 5$: 1302 dimers.

$R(K, r_x) \approx 1.00$, $|\rho| \approx 0.03$. $R(K, w_{\text{CHM}}) \approx 0.16$, $\rho \approx -0.987$. r_x is spatially flat. $\langle r_z \rangle = 0$.

C2 FAIL. Auditor REFUTED. The Bloch rewrite does not describe K . CHM does. This covering is dense enough to end that argument. Curvature as a Bloch axis was not claimed and is still a guess.

1 Numbers

	dimers	R_{Bloch}	R_{CHM}	ρ_{Bloch}
$N = 16, R = 5$	1302	0.999	0.162	0.034
$N = 12, R = 4$	624	0.999	0.156	-0.033
audit $N = 14, R = 5$	1302	1.000	0.163	0.031

Admission. *I ran the Bloch test I had named and I covered the ball. Bloch lost. I did not identify r_z with curvature, and I did not write that the guess was a habitat.*